

1M1B × IBM SKILLSBUILD AI FOR SUSTAINABILITY INTERNSHIP

MealCycle AI

Smart Canteen Decision Support & Food Waste Prevention System

Human-in-the-Loop AI for SDG 2 (Zero Hunger) through efficient use of food resources, and SDG 13 (Climate Action) through reduced resource and energy waste from overproduction.

 AI • Sustainability • Waste Prevention

Primary SDG: Responsible Consumption & Production

MealCycle AI targets the institutional food-service segment of Target 12.3 — canteens are a consumption node where over-preparation directly causes avoidable waste.

Secondary alignment: SDG 2 (Zero Hunger) through efficient use of food resources, and SDG 13 (Climate Action) through reduced resource and energy waste from overproduction.



MealCycle AI: Decision Support, Not Automation

Core Philosophy: Decision Support, Not Automation (Human-in-the-Loop)



Demand Forecasting Engine

Linear Regression ML Model



Natural Language Shift Note Extractor

AI Prompt Engineering4



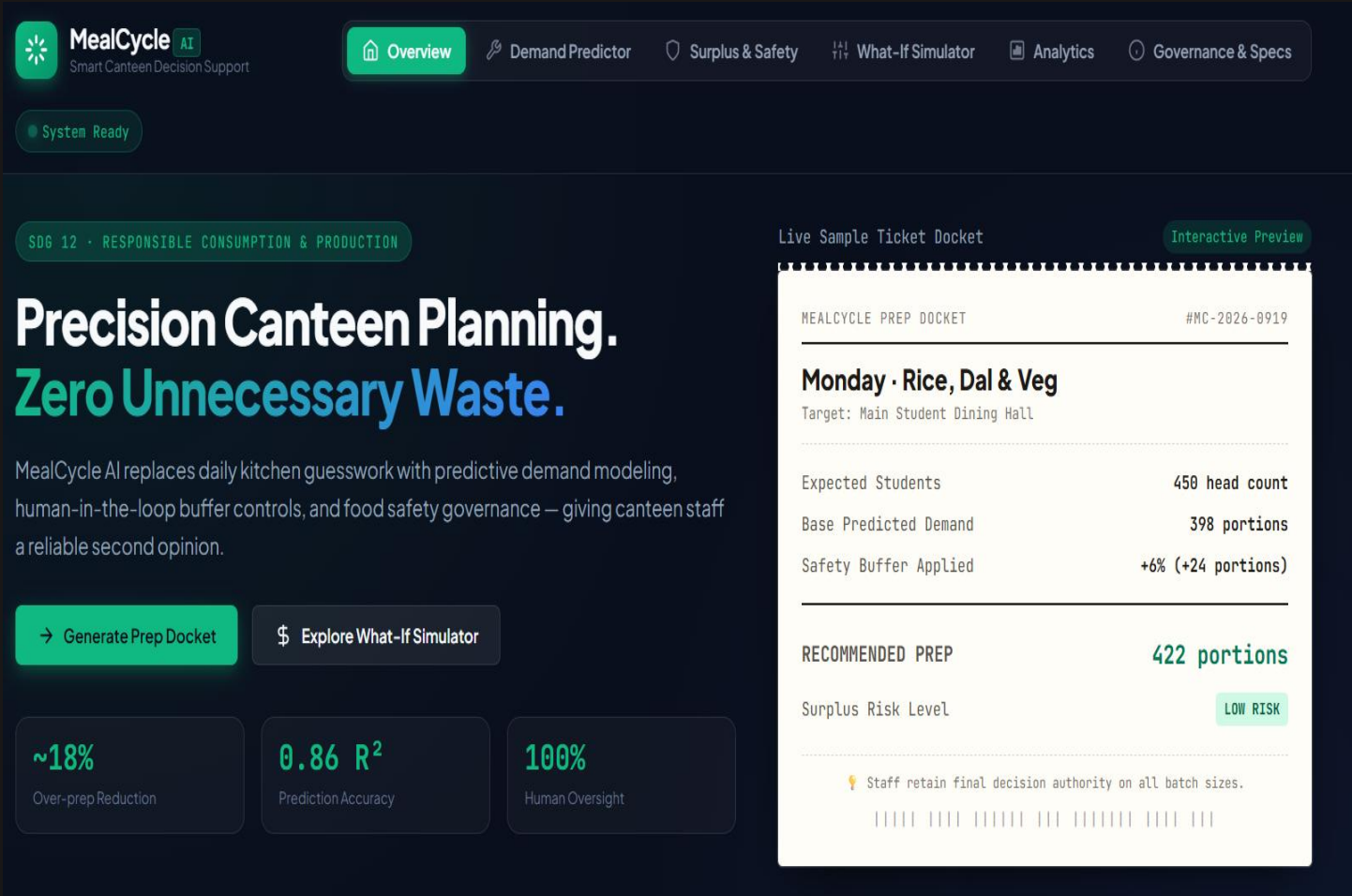
Grounded Food Safety RAG Assistant

Vector Policy Lookup



What-If Impact Simulator

\$ Cost & CO2e Footprint Savings





SUSTAINABILITY

The Problem: Institutional Food Waste Crisis

Institutional canteens over-prepare food by 15-25% daily due to manual guesswork, causing massive financial loss and organic food waste.

UN SDG 12.3 Alignment: Halving per capita food waste by 2030

15-25%

Daily Over-Preparation

of food in institutional canteens

0.8 kg

Carbon Impact

of CO2e carbon emissions per 1 portion of wasted food



5. Final Supervisor Review & Batch Authorization

Tomorrow's Operational Context

EXPECTED STUDENTS / ATTENDEES

420

DAY OF WEEK

Wednesday

▼

MENU TYPE

Regular Rotation

▼

CONTEXT MODIFIERS

☐

CAMPUS EVENT (+EVENT DEMAND)

☐

EXAM SEASON (-TURNOUT)

☐

HOLIDAY / PARTIAL DAY

HUMAN SAFETY BUFFER: +6%

0% (Exact)

+6% (Standard)

+20% (High Safety)

MEALCYCLE PREP DOCKET

#MC-2026-5351

Wednesday · Regular Rotation

Generated: Fri, Sep 18 · Target: Main Hall

Expected Students

Model Forecast Demand

Safety Buffer (+6%)

420 head count

≈ 387 portions

+23 portions

RECOMMENDED PREP

410 portions

Surplus Risk Rating

LOW RISK

Human Decision Support – Staff authorize final kitchen batches.

||||| ||||| ||||||| ||| ||||||| |||||

Print Kitchen Docket

Copy Text

ML Model Performance: Multiple Linear Regression

0.864

R² Score

25.04

MAE (portions)

28.91

RMSE (portions)

Learned Model Coefficients:

Tomorrow's Operational Context

EXPECTED STUDENTS / ATTENDEES

525

DAY OF WEEK

Friday

MENU TYPE

Rice + Dal + Vegetables

CONTEXT MODIFIERS

☒ CAMPUS EVENT (+EVENT DEMAND)

☐ EXAM SEASON (-TURNOUT)

☐ HOLIDAY / PARTIAL DAY

HUMAN SAFETY BUFFER: +10%

MEALCYCLE PREP DOCKET#MC-2026-4404

Friday · Rice+Dal+VegGenerated: Fri, Sep 18 · Target: Main Hall

Expected Students525 head countModel Forecast Demand≈ 549 portionsSafety Buffer (+10%) +55 portions

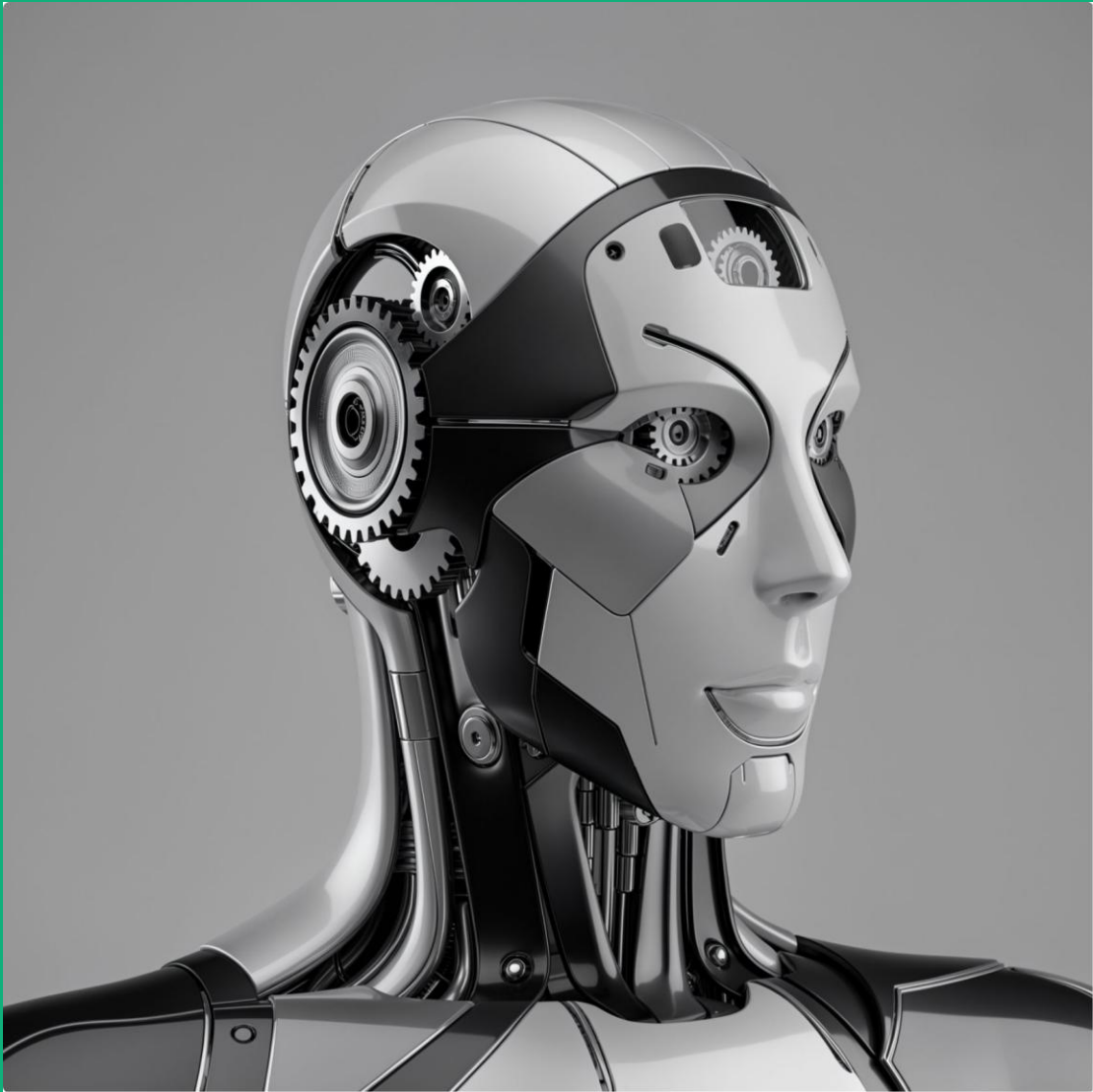
RECOMMENDED PREP604 portions

Surplus Risk RatingMEDIUM RISK

Human Decision Support – Staff authorize final kitchen batches.

Print Kitchen DocketCopy Text

AI-Powered Extraction & Safety Guardrails



AI Note Parser



RAG Policy Assistant

Interactive Impact Modeling: What-If Scenarios

Interactive Dual Scenario Modeling: Test turnout (-40% to +40%) and lean cooking (-30% to +30%) before cooking.

Turnout Variability

Adjust predicted attendance by -40% to +40%

Lean Cooking Adjustment

Reduce portion size by -30% to +30% for lean cooking

Quantifiable Impact Metrics:

\$2.50

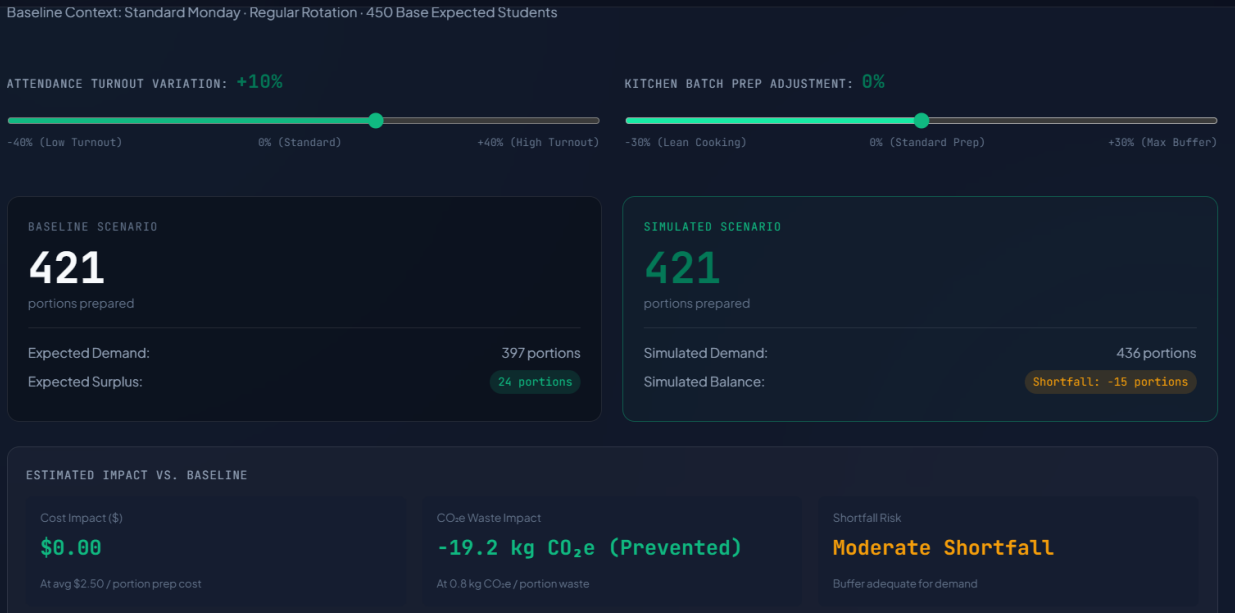
Financial Savings

Calculated at average preparation cost per portion

0.8 kg

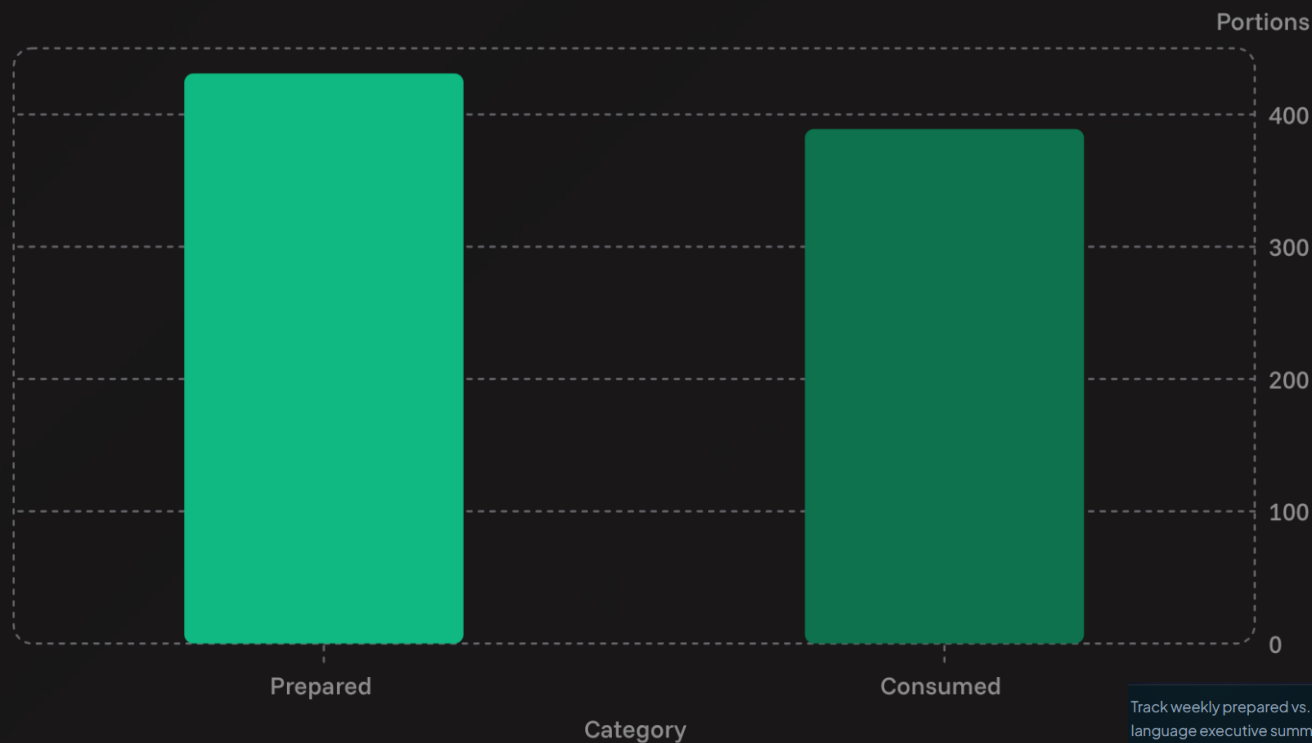
Carbon Waste Savings

CO2e emissions prevented per portion saved



Executive Dashboard: Weekly Tracking & Insights

Weekly Tracking Dashboard: Prepared vs. Consumed



This chart visually represents the portions of food prepared versus consumed during the week, highlighting areas for optimization.

Key Metrics

414	Avg Daily Students
431	Prepared
389	Consumed
42	Surplus

1-Click Executive Digest: Automated plain-language weekly summary for institutional leadership

Track weekly prepared vs. consumed ratios, analyze waste distribution, and generate automated plain-language executive summaries.

414	431	389	42
Avg Daily Attendance	Avg Daily Prepared	Avg Daily Consumed	Avg Daily Surplus
↑ 3.2% vs last wk	Stable	↑ 4.1% efficiency	↓ 14% reduction



Responsible AI: Fairness, Transparency & Human Control

Content with 4 key pillars:



Fairness & Equity

Aggregated headcount modeling only (Zero individual student tracking or profiling)



Human-in-the-Loop

AI outputs are recommendations; authorized staff retain sole decision authority



Transparency

Explicit breakdown of base forecast vs. safety buffer on every ticket



Data Privacy

Zero PII collected or stored

● System Ready

Governance & System Methodology

MealCycle AI adheres to strict Responsible AI guidelines (SDG 12), prioritizing human oversight, data privacy, and ethical decision support.



Fairness & Equity

Models focus on aggregated daily student headcounts. Individual consumption profiling is explicitly prohibited to prevent bias.



Transparency

Every generated prep ticket displays explicit breakdown parameters — baseline model forecast, day weights, and applied safety buffer.



Human-in-the-Loop

AI outputs are strictly recommendations. Authorized kitchen managers maintain sole authority over final prep quantities.



Data Privacy

Operates entirely on aggregated institutional attendance totals and anonymized meal counts. Zero personal identifiable information (PII) processed.



Food Safety Boundary

The AI system never independently certifies food as safe/unsafe for redistribution. Safety decisions strictly follow retrieved institution protocols.



Sustainability Target

Aligned with UN Sustainable Development Goal 12.3: Halving global food waste at retail and consumer levels by 2030.

Conclusion & Future Roadmap

Conclusion: Demonstrates accessible AI delivering immediate, measurable SDG 12 impact.

Future Roadmap

1

Smart card attendance gate
integration

2

Weather API automation

3

Multi-canteen rollout

Thank You